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Original Article: Clinical Investigation

Medical mushrooms used for biochemical failure after radical treatment for prostate cancer: An open-label study

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Objective: The aim of this study was to assess the efficacy and safety of two different types of medical mushrooms in patients with prostate cancer in Japan.

Methods: Patients with biochemical failure after radical treatment for non-metastasized prostate cancer were enrolled in this open-label study. For 6 months they ingested one of the two following supplements: *Senseiro*, containing extracts from the *Agaricus blazei* Murill mushroom; and *Rokkaku Reishi*, containing the *Ganoderma lucidum* mushroom. Levels of serum prostate-specific antigen (PSA) level and PSA doubling time were examined before and after study entry to assess the impact of these supplements on disease progression. The primary end-point of this study was partial response rate (50% or more decrease of serum PSA). Hormonal status, represented by serum testosterone levels, and toxicity were also assessed.

Results: A total of 51 patients were enrolled following radical prostatectomy. Forty-seven completed the protocol and could be assessed. Thirty-two patients received *Senseiro* and the remaining 15 received *Rokkaku Reishi*. No partial response in terms of PSA was observed. Alteration of PSA doubling time did not correlate with that of serum testosterone levels. Serious adverse effects were not observed.

Conclusions: No significant anticancer effects were observed with the intake of these two medical mushrooms.

Key words: complementary medicine, prostate caner, prostatectomy, prostate-specific antigen, radiation.

Introduction

Prostate cancer is the most common male cancer in North America, and the fourth most common cancer in Japanese men. As prostate cancer generally does not have any aggressive biological features, patients may suffer from a long period of cancer burden. The high prevalence of this cancer and the long periods involved often lead patients to become interested in complementary and alternative medicine (CAM).

CAM is defined as 'a group of diverse medical and health care systems, practices, and products that are not presently considered to be part of conventional medicine' by the National Center for Complementary and Alternative Medicine (NCCAM) in the USA.[1] CAM encompasses various forms of medicine, such as dietary supplements, megadose vitamins, herbal preparations, special teas, acupuncture, massage therapy, magnet therapy, spiritual healing, and meditation. Since the study by Eisenberg *et al*., patients, medical doctors and even the governments of several countries have become gradually more interested in CAM.[2] The fiscal year budget of the NCCAM in the USA was $121 577 000 in 2008. This large budget contrasts sharply with the scarcity of evidence for the efficiency of CAM in various diseases, including malignant ones. A high prevalence of CAM use by cancer patients has also been reported in Japan; up to 45% of cancer patients were reported to have used CAM in order to prevent disease development and recurrence and/or to aid treatment.[3,4]

The most prevalent types of CAM used by Japanese patients with prostate caner are extracts from the *Agaricus blazei* Murill (ABM) mushroom.[4,5] Yu *et al*. recently reported that, *in vitro*, extracts from ABM significantly inhibited cell proliferation in both androgen-dependent and androgen-independent prostate cancer cell lines by enhancement of apoptotic pathways and that, *in vivo*, tumor xenografts from ABM-fed mice showed decreased proliferating-cell nuclear antigen-positive cells and reduced tumor microvessel density.[6] Despite the popularity of dietary CAM, especially medical mushrooms including ABM, and

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Table 1 Patient characteristics

		Senseiro		*Rokkaku Reishi*	
No. of Patients		34		17	
Age (years)	<60	4	(12)	2	(12)
	60–69	10	(29)	5	(29)
	70–80	20	(59)	10	(59)
Clinical stage	B0	10	(29)	3	(18)
	B1	8	(24)	4	(24)
	B2	10	(29)	7	(41)
	C1	6	(18)	3	(18)
	C2	0	(0)	0	(0)
Performance status	0	33	(97)	17	(100)
	1–	1	(3)	0	(0)
Complications	No	16	(47)	11	(65)
	Yes	18	(53)	6	(35)
History of CAM use	No	31	(91)	14	(82)
	Yes	3	(9)	3	(18)

CAM, complementary and alternative medicine.

the recent accumulation of evidence on their anticancer characteristics from basic research, there has been little clinical research regarding their anticancer effects.[7] Thus, it would be highly beneficial for both urologists and patients with prostate cancer to elucidate the clinical efficacy and safety of some types of dietary CAM.

The purpose of this study was to investigate the efficacy and safety of some types of medical mushrooms, including extracts from ABM, which are currently used by patients with prostate cancer in Western countries. Randomized trials are required to test whether or not such medical mushrooms have beneficial effects. However, it is very difficult to divide patients into two or three groups by randomization, because such medical mushrooms are easily accessible through supplement shops. Based on this reasoning, in this study patients were enrolled into two cohorts depending on their voluntary decisions to take either form of mushroom extract.

Methods

This study was approved by the institutional review boards (the ethical committees) of the Kyoto University Graduate School of Medicine, Kurashiki Central Hospital, Tenri Yorozu Hospital, and Akita University Faculty of Medicine. All patients provided written informed consent.

Patients with non-metastasized prostate cancer, who had experienced a biochemical recurrence after treatment with intent to cure, including radical prostatectomy or external-beam radiotherapy without adjuvant therapy, were eligible for this study. Biochemical recurrence was defined as the value of serum prostate-specific antigen (PSA) increasing to >0.1 ng/mL following radical prostatectomy, or three consecutive increases in PSA value (after the nadir has been reached) with the third increase being >1.0 ng/mL after external-beam radiotherapy.[8]

Criteria for inclusion in the study were as follows: (i) age under 80 years old at the time of study entry; (ii) no clinical recurrence detected by thoracic and abdominal computed tomography examination; (iii) no definite anticancer treatment after radical therapy, including hormonal agents, radiation therapy, chemotherapy, or investigational treatment; (iv) no severe complications, including heart disease, diabetes mellitus, or other malignant disease; and (v) serum transaminase (aspartate aminotransferase and alanine aminotransferase) levels within twice the upper limits of normal ranges. Prior androgen deprivation was permitted in the neoadjuvant setting before radical therapy.

Patients were recruited from December 2003 to November 2006. Enrollment and data management were performed in an independent data center at Translational Research Center, Kyoto University Hospital.

Medical mushrooms studied

In this study, two types of medical mushroom extract were examined. One was a liquid dietary supplement, called *Senseiro* (Kyowa Wellness, Tokyo, Japan), containing a standardized extract of the ABM mushroom. The other was also a Japanese dietary supplement, called *Rokkaku Reishi* (Suntory Holdings, Osaka, Japan). This product is made from the antlered form of the mushroom *Ganoderma lucidum*. Japanese regulations require that a use-by date of

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less than 2 years appears on both of these products, because they count as foodstuffs. Due to these regulations, samples of *Senseiro* were taken from two lots; one was provided at study entry and the second after 18 months. Patients could choose their preferred product.

Study protocols

Patients received three packs of either product per day for 6 months, unless adverse effects occurred. Patients were asked to record their daily intake and to submit the records. Daily intake records were checked by physicians, and an intake compliance of less than 80% resulted in exclusion from the study.

Serum PSA level, complete blood count, and serum chemistry including transaminases were examined at study entry and every 2 months thereafter. These examinations, except for PSA, were also performed 2 weeks after entry in order to check for acute toxicity. Serum testosterone levels were examined at study entry and at months 2 and 6, and thereafter every 2 months. For the assessment of PSA response, we evaluated patients only after radical prostatectomy. Patients were questioned about adverse effects such as allergic reaction or fatigue, and these, if any, were recorded according to the Japanese version of the National Cancer Institute Common Toxicity Criteria, version 2.[9] For the assessment of safety, we evaluated all patients enrolled in the study.

After the initial 6 months protocol, patients were allowed to decide whether to continue the study for a further 2 years. Patients therefore received their chosen CAM agent for a maximum of 30 months. During this extended period, checks for compliance and monitoring of serum PSA levels were continued.

The primary end-point of this study was the effects of the two CAM agents on PSA levels in patients. We used the criteria of PSA recurrence, represented by the PSA partial response rate, defined as the proportion of patients who achieved a greater than 50% reduction in PSA value 6 months after administration of CAM compared with the baseline PSA value at study entry. We also evaluated the PSA doubling time (PSADT), comparing the value for study entry to 6 months with that taken before study entry. We calculated PSADT as the natural log of 2 (0.693) divided by the slope of the correlation between the log of PSA and time of PSA measurement.[10] The secondary end-points of this study were: (i) the safety profiles of the agents, in particular the incidence of liver toxicity; and (ii) the effect on testosterone levels.

Statistical considerations

We took a PSA partial response of <1% under no treatment or in the absence of an effect of CAM as our null hypothesis, and the alternative hypothesis was a PSA partial response rate of 10%, taken to be a clinically significant level in the presence of an effect of CAM. We planned a group sequential design that involves performing interim analyses whenever the responses from an additional 10 patients were obtained for each CAM agent. We determined that we required 40 patients to achieve a type I error rate of 5% and a statistical power of 90% for each CAM agent. For comparison of PSADT values before and after study entry, PSA slopes, calculated as 1/PSADT, were used. A Wilcoxon signed-rank test was used for comparison of PSA slope, and Spearman's rank correlation coefficient was used to examine correlations between alterations in PSADT and testosterone levels. $P < 0.05$ was considered to be statistically significant.

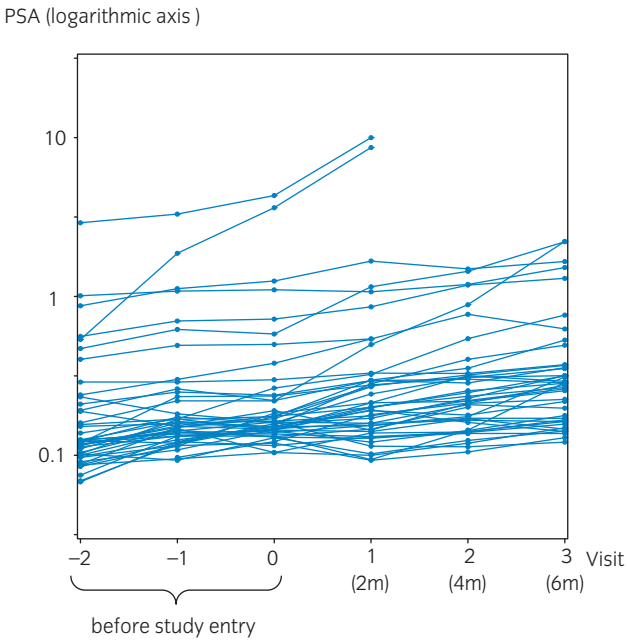

Fig. 1 Alterations in serum prostate-specific antigen (PSA) values before and after study entry.

Results

Patient characteristics

The total number of patients enrolled in this study was 51, and 34 and 17 of them selected *Senseiro* and *Rokkaku Reishi*, respectively. Of all patients, 47 entered the study after radical prostatectomy (32 and 15 of those selecting *Senseiro* and *Rokkaku Reishi*, respectively), and four entered the study after radiotherapy (two selecting *Senseiro* and two selecting *Rokkaku Reishi*). Detailed patient characteristics are shown in Table 1. Of all patients, 88.2% had no history of any CAM usage before study entry. Although we attempted initially to recruit 40 patients for each group, we

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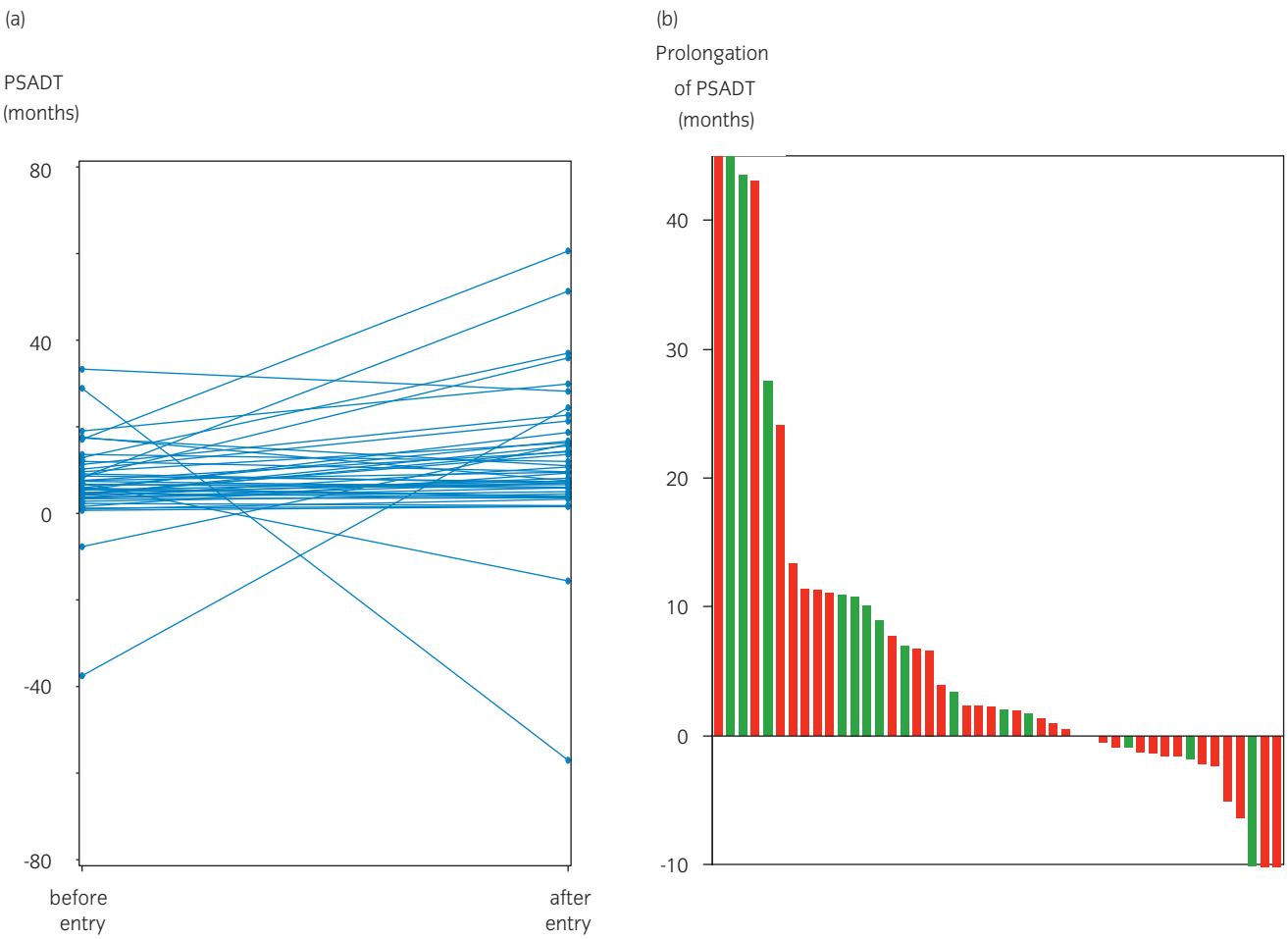

Fig. 2 (a) Comparison of prostate-specific antigen doubling time (PSADT) before and after study entry. (b) A waterfall plot showing alterations of PSADT before and after study entry. The values of PSADT alteration are PSADT after study entry minus that before study entry. Red bars indicate patients taking *Senseiro*, and green bars indicate those taking *Rokkaku Reishi*. The leftmost two bars show the alterations from positive to negative values of PSADT, and the rightmost two bars show the alterations from negative to positive values of PSADT.

decided to cease recruitment at the time of the interim analyses. No patient showed a partial response of serum PSA level at the interim analyses (see below), and we judged that further recruitment of patients would not be meaningful for this study. Patients who selected *Senseiro* or *Rokkaku Reishi* remained in the study for an average of 6 and 7 months, respectively, after the initial 6 months. Patients continued to take *Senseiro* for 2–32 months, and *Rokkaku Reishi* for 6–25 months. No patients left the study due to any serious adverse effects, although three patients did leave following a rapid rise in serum PSA.

Impacts on PSA levels

The response of PSA values was analyzed in the 47 patients undergoing prostatectomy. The observed partial response rate was 0% for both CAM agents. Figure 1 shows the actual alteration of PSA levels before and after study entry. Figure 2 shows the comparison of PSADT before and after study entry. For *Senseiro*, the PSADT at 6 months was significantly longer than that at study entry ($P = 0.03$; median, 8.3–9.6 months). We also observed this tendency for *Rokkaku Reishi* ($P = 0.02$; median, 5.8–10.7 months).

After the formal study period, 23 patients (13 taking *Senseiro*, and 10 taking *Rokkaku Reishi*) continued to ingest CAM agents for up to 12 months from study entry. At 12 months after entry, the PSADT of the *Senseiro* group was not significantly prolonged ($P = 0.19$), and that of the *Rokkaku Reishi* group was marginally prolonged ($P = 0.06$) compared with values before entry.

Endocrine effects

At the 6-month analysis a reduction in testosterone level of at least 20% compared with baseline testosterone level at study entry was observed for 30% of patients using *Senseiro*

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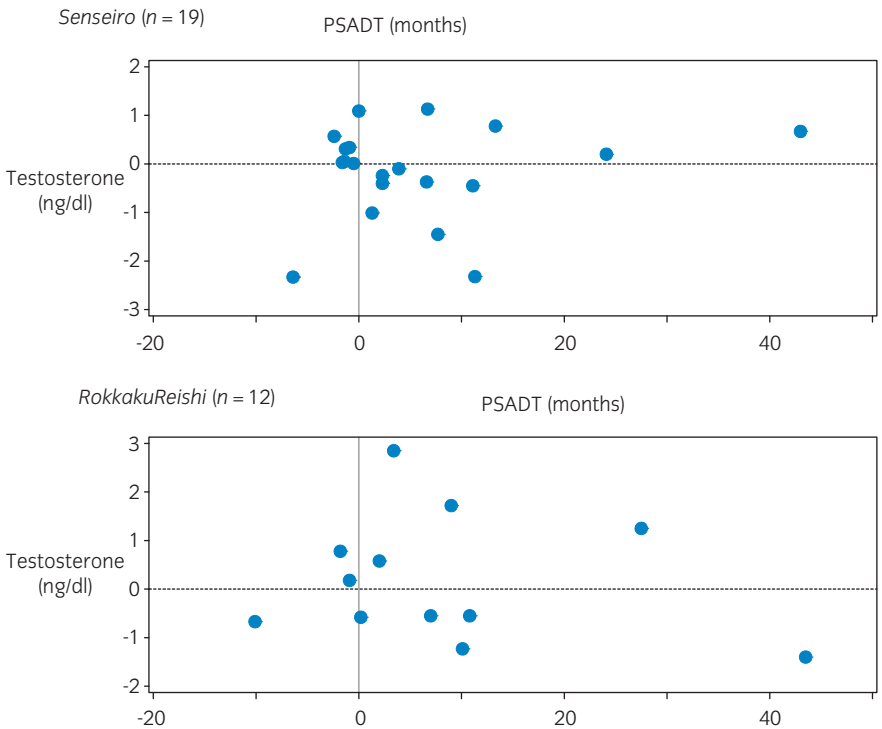

Fig. 3 Correlation between alterations in testosterone levels and those of prostate-specific antigen doubling time (PSADT). Spearman's correlation coefficients, *R*, were 0.01 and 0.05, and *P*-values were 0.97 and 0.87, in the *Senseiro* and *Rokkaku Reishi* groups, respectively. The data from four patients whose PSADT values were negative before or after study entry were excluded.

and 12% of patients using *Rokkaku Reishi*, although no patients dropped outside the normal range of testosterone. Furthermore, there was no correlation between alterations in testosterone levels and those of PSADT (Fig. 3, $R = 0.01$ and $P = 0.97$ for *Senseiro*; $R = 0.05$ and $P = 0.87$ for *Rokkaku Reishi*). This result does not support the hypothesis that there is a correlation between the drop in testosterone levels and prolongation of PSADT.

Toxicity

As shown in Table 2, some patients experienced mild adverse effects, but no patients had any serious adverse effects. No patient who continued the study after the initial 6-month protocol suffered any adverse effects.

Discussion

It is accepted that the incidence and the associated mortality rates of some types of cancer, including prostate cancer, are affected not only by genetics but also by some environmental characteristics, especially daily food intake.[11] There have been several clinical trials of dietary agents for hormone-refractory prostate cancer.[7] For example, Ornish *et al*. reported a positive impact on prostate cancer of environmental changes.[12] They randomized patients with early-stage, non-aggressive and androgen-dependent prostate cancer into two groups: an experimental group who changed their lifestyle to a vegan diet and moderate aerobic exercise, and a control group without any intervention. In the 12 months following diagnosis, serum PSA decreased by 4% in the experimental group, whereas it increased by 6% in the control group.[12] By the 2-year follow up, the rate of medical intervention was significantly lower in the former group than in the latter.[13] This study suggested that environmental factors including diet could influence the progression of prostate cancer.[12,13]

With respect to medical mushrooms, only one clinical study has so far been reported on the effectiveness of shiitake mushroom extract for progressing prostate cancer. Of the 61 men assessed, there were no PSA responses in this study. Regarding ABM, despite several previous studies (including that shown in the *Physicians' Desk Reference*) showing no significant direct effect on cancer cell growth, Yu *et al*. reported that ABM had a direct anticancer effect on prostate cancer cells, as mentioned in the introduction to this paper.[6] Furthermore, we have seen one case of hormone-refractory prostate cancer for which the serum PSA level had decreased markedly after ABM ingestion. Thus, our primary end-point was to elucidate the degree of anticancer effects of medical mushrooms, including ABM, in a clinical setting.

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Table 2 Adverse events

Senseiro (*n* = 33) Adverse events		Grade 1	2	3	4	5
Blood/Bone marrow	Other (leukocytosis)	1 (3)	0	0	0	0
Constitutional symptoms	Fatigue	5 (15)	0	0	0	0
Dermatology/Skin	Rash	0	1 (3)	0	0	0
Gastrointestinal	Diarrhea	2 (6)	0	0	0	0
	Dyspepsia/heartburn	1 (3)	0	0	0	0
Hepatic	SGOT (AST)	1 (3)	1 (3)	0	0	0
	SGPT (ALT)	3 (9)	1 (3)	0	0	0
	γ-GTP	1 (3)	0	0	0	0
Musculoskeletal	Other	1 (3)	0	0	0	0
Pain	Myalgia	1 (3)	0	0	0	0
Pulmonary	Pneumonitits/ pulmonary infiltrates	1 (3)	0	0	0	0
	Voice change/strider	1 (3)	0	0	0	0
Rokkaku Reishi (*n* = 17) Adverse events		Grade 1	2	3	4	5
Blood/Bone marrow	Hemoglobin	0	2 (12)	0	0	0
	Other (leukocytosis)	1 (6)	0	0	0	0
Dermatology/Skin	Rash	0	1 (6)	0	0	0
Gastrointestinal	Constipation	2 (12)	0	0	0	0
	Dyspepsia/heartburn	1 (6)	0	0	0	0

Percentages are in parentheses. ALT, alanine aminotransferase; AST, aspartate aminotransferase; GTP, glutamyl transpeptidase; SGOT, serum glutamate oxaloacetate transaminase; SGPT, serum glutamate pyruvate transaminase.

Our study showed no direct anticancer effect of the medical mushrooms examined because no patient obtained a partial response in serum PSA level. This observation is very important and suggests that physicians and patients should not expect objectively constant or steady disease improvement when using these compounds. Although our study also showed a slight but statistically significant delay of PSADT by these compounds in some patients, we should interpret this finding carefully. We did not have a placebo-control group, and the PSA levels of our patients were too low to be reliable for calculation of PSADT. However, these supplements may have some mild impact on prostate cancer by an indirect mechanism. In support of this idea, Ahn *et al*. reported in their study of patients with gynecological cancer that natural killer cell activity was significantly higher in the ABM-treated group and that chemotherapy side-effects, including appetite loss, alopecia, emotional stability, and general weakness, were improved by ABM treatment.[14] Because the modulation of immunoreactivity by ABM has been well examined in studies *in vivo*,[15] enhancement of the immune system and/or improvement of physiological and psychological conditions might have induced a mild delay in PSADT in our study. This study cannot clarify such impacts, but it should not be overlooked that these effects may indirectly influence cancer cell growth.

This study also examined and evaluated the safety profile of two specific CAM products. They caused almost no adverse events, except for mild hepatic damage and mild gastrointestinal disturbance in a very small population. In this study an influence of these products on testosterone could be identified, but there was no conclusive effect. Although some patients showed a decrease in their testosterone levels, none dropped outside normal range. As there is no evidence of any correlation between PSADT prolongation and a decrease in testosterone levels, it can be inferred that these decreases in testosterone are most likely normal fluctuations.

Our study has several limitations. First, we did not use a placebo-control group. Therefore, we should interpret our results carefully, especially the delay of PSADT, as mentioned above. Whereas a placebo-controlled randomized trial would have been better, strict prohibition of use of dietary CAM, which can be easily obtained by patients, would be quite difficult. Second, our study population was relatively small. Although we attempted to recruit 40 patients in each group initially, we decided to cease

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recruitment because no objective response was observed at the interim analyses. Third, the definition of biochemical failure after prostatectomy is not generally approved at present. Although a PSA value of 0.2 ng/mL after prostatectomy is currently recommended,[16] there were various definitions of PSA failure. Because an ultra-sensitive assay for PSA is used widely in Japan, 0.1 ng/mL is not an unreasonable cut-off value. Similarly, the guidelines for calculation of PSADT have been developed recently, and they mention that PSA values in the calculation should be 0.2 ng/mL or greater.[17] Because our study included many PSA values below 0.2 ng/mL, the results from PSADT measurement should again be interpreted carefully. Finally, there may be literally hundreds of unique compounds in a herbal mixture, making it difficult to identify the active agent and to control inter- and intralot variability. In this study we attempted to minimize the effects of interlot variability by using the minimum number of lots possible. However, it cannot be ignored that even within this study some variability due to different lots being used may have occurred. Therefore, while it is necessary for further tests to be conducted, it is expected that future tests will show slightly different results due to inter- and intralot variability. Despite these limitations, we believe that our study presents quite valuable information regarding the use of medical mushrooms to both urologists and patients with prostate cancer.

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